Lecture 13

Spin groups and projective representations

Objectives. *A 2π rotation is the identity in* $\mathrm{SO}(3)$. *On an electron's state vector it acts by* -1. *We explain this sign using the double cover* $\mathrm{Spin}(n) \to \mathrm{SO}(n)$. *We then study actions on rays and their lifts to vectors.*

By the end of this lecture you will be able to:

- For $n \geq 2$, construct $\mathrm{Spin}(n)$ and prove that $\rho\colon\ \mathrm{Spin}(n) \to \mathrm{SO}(n)$ is surjective with kernel $\{\pm 1\}$.
- Identify $\mathrm{Spin}(3)$ with $\mathrm{SU}(2)$ and decide which V_m descend to $\mathrm{SO}(3)$.
- Define projective unitary representations, multipliers and central extensions, and lift finite-dimensional projective representations of $\mathrm{SO}(3)$ to $\mathrm{SU}(2)$.
- Use $\mathbb{R}^2$ to explain why a simply connected group can have a nontrivial multiplier.

13.1 Clifford algebras and spin groups

A rotation is a product of an even number of reflections. A reflection is determined by a unit vector u, up to the sign of u. We want an algebra in which conjugation by u gives the reflection in $u^{\perp}$ up to sign. For a product of an even number of unit vectors the signs cancel. The Clifford algebra has this property. For $n = 2$, Exercise 13.1 shows the result on a circle.

Fix $n \geq 2$ and an orthonormal basis $e_1, \dots, e_n$ of Euclidean $\mathbb{R}^n$. Recall from Lie Algebras that the *Clifford algebra* $Cl(n)$ is the real unital associative algebra generated by $\mathbb{R}^n$ subject to $v^2 = -|v|^2$. Replacing v by $u + v$ gives

$$uv + vu = -2\langle u, v\rangle \qquad (u, v \in \mathbb{R}^n). \tag{13.1}$$

This is the prerequisite convention $uv + vu = 2q(u, v)$ with the negative definite symmetric bilinear form $q(u, v) = -\langle u, v\rangle$.

The products $e_I = e_{i_1} \cdots e_{i_k}$ with $i_1 < \cdots < i_k$ form a basis of $Cl(n)$. The *even part* $Cl^0(n)$ is the span of the e_I with $|I|$ even. The *transpose* is the linear map $x \mapsto x^t$ with $(v_1 \cdots v_k)^t = v_k \cdots v_1$. The complexification $Cl(n)_{\mathbb{C}}$ has an irreducible *spinor module* S of dimension $2^{\lfloor n/2 \rfloor}$.

The units $Cl(n)^{\times}$ form an open set and a Lie group. A unit vector u has $u^2 = -1$ and $u^{-1} = -u$. Let $r_u(v) = v - 2\langle u, v\rangle u$ be reflection in $u^{\perp}$.

Definition 13.1. $\mathrm{Spin}(n)$ is the set of all products $u_1u_2\cdots u_{2k}$ with $k \geq 0$ and unit vectors $u_i \in \mathbb{R}^n$.

Lemma 13.2. *(a)* $\mathrm{Spin}(n)$ *is a subgroup of* $Cl^0(n)^\times$, *and* $x^{-1} = x^t$ *for* $x \in \mathrm{Spin}(n)$.
(b) For a unit vector u *and* $v \in \mathbb{R}^n$ *we have* $uvu^{-1} = -r_u(v)$.

Proof. (a) Even products of vectors lie in $Cl^0(n)$, and the set is closed under products. Let $x = u_1\cdots u_{2k}$, so $x^t = u_{2k}\cdots u_1$. In

$$x^t x = u_{2k}\cdots u_2\,(u_1u_1)\,u_2\cdots u_{2k}$$

the middle factor is the scalar -1. Next the two factors u_2 become adjacent and give $u_2u_2 = -1$, and so on. Hence $x^tx = (-1)^{2k} = 1$, and in the same way $xx^t = 1$. Since $x^t \in \mathrm{Spin}(n)$, the set is a subgroup.

(b) By the anticommutation relation (13.1), $uv = -vu - 2\langle u, v\rangle$. Using also $u^{-1} = -u$ and $u^2 = -1$, we get

$$uvu^{-1} = -uvu = (vu + 2\langle u, v\rangle)u = -v + 2\langle u, v\rangle u = -r_u(v). \qquad \blacksquare$$

13.2 The double covering of the rotation group

For $x \in \mathrm{Spin}(n)$ and $v \in \mathbb{R}^n$ put $\rho(x)v = xvx^{-1}$, an element of $Cl(n)$. Write $x = u_1\cdots u_{2k}$ with unit vectors $u_i \in \mathbb{R}^n$. Apply the conjugation formula of Lemma 13.2(b) once for each of the $2k$ factors of x. The $2k$ signs cancel, and we get

$$\rho(x)v = xvx^{-1} = r_{u_1}\circ\cdots\circ r_{u_{2k}}(v) \qquad (v \in \mathbb{R}^n). \tag{13.2}$$

So $\rho(x)$ maps $\mathbb{R}^n$ to itself, and it is a product of an even number of reflections. Hence $\rho(x) \in \mathrm{SO}(n)$, and ρ is a map from $\mathrm{Spin}(n)$ to $\mathrm{SO}(n)$.

The formula (13.2) explains the double cover. A reflection r_u does not change if we replace u by $-u$, but the element u of $Cl(n)$ does change. So x and $-x$ give the same rotation.

Two mirrors show this in a plane. Let e_a and e_b be orthonormal, let $u = \cos t\, e_a + \sin t\, e_b$, and put $x = -ue_a = u(-e_a)$. By (13.2), $\rho(x) = r_u \circ r_{e_a}$, so we reflect first in $e_a^\perp$ and then in $u^\perp$. These two mirrors meet at angle t. Example 13.5 below shows that the composition is the rotation through $2t$ from e_a towards e_b. Figure 13.1 draws the two reflections. In its second row, t runs from 0 to π. The mirror of u returns to its start, the rotation runs once around, and the product x runs from 1 to -1. At $t = \pi$ we have $u = -e_a$, with the same mirror and the same rotation, but with product -1.

Lemma 13.3 (Cartan–Dieudonné, weak form)**.** *Every* $g \in \mathrm{O}(n)$ *is a product of at most* n *reflections. If* $\det g = 1$, *the number of reflections is even.*

Proof. Induct on n. For $n = 1$ we have $\mathrm{O}(1) = \{1, r_{e_1}\}$. Let $n \geq 2$. If $ge_1 = e_1$, then g preserves $e_1^\perp \cong \mathbb{R}^{n-1}$. By induction $g|_{e_1^\perp}$ is a product of at most $n-1$ reflections of $e_1^\perp$, and each extends

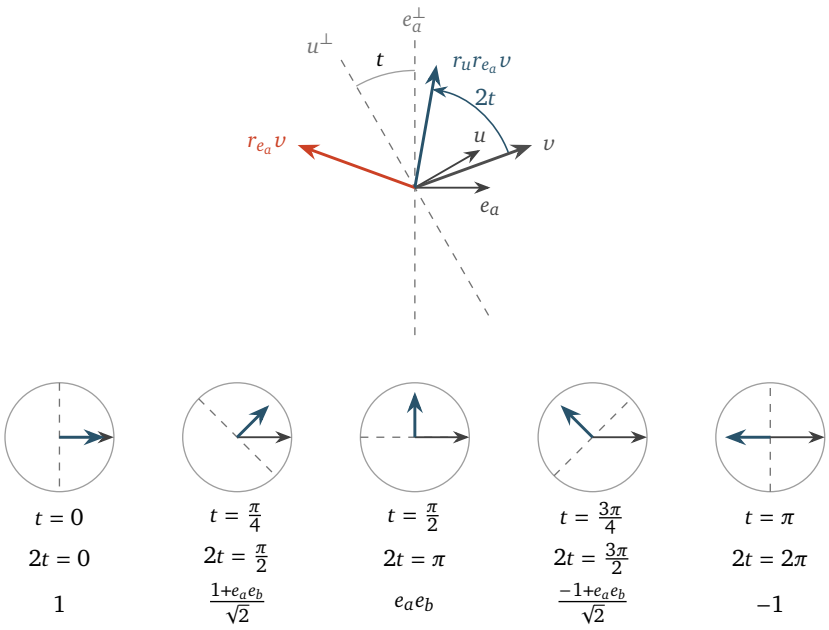


Figure 13.1: Two mirrors in the plane of e_a and e_b. The first panel has $t = \pi/6$: a vector v at angle $\pi/9$ goes to $r_{e_a} v$, and then to $r_u r_{e_a} v$, which is v turned through $2t$. The second row shows e_a, the vector u in colour, and the dashed mirror $u^\perp$, for five values of t. Beneath each are the rotation angle $2t$ and the product $x = -ue_a$.

to a reflection of $\mathbb{R}^n$ fixing e_1. If $a = ge_1 \neq e_1$, put $u = (a - e_1)/|a - e_1|$. Since $|a| = 1$, we have $\langle a - e_1, a\rangle = \frac{1}{2}|a - e_1|^2$. Therefore

$$r_u(a) = a - \frac{2\langle a - e_1, a\rangle}{|a - e_1|^2}(a - e_1) = e_1.$$

So $r_u g$ fixes e_1, and $g = r_u(r_u g)$ uses at most n reflections. Each reflection has determinant -1, which gives the parity. ∎

Theorem 13.4. *Let $n \geq 2$.*

(a) $\rho\colon \mathrm{Spin}(n) \to \mathrm{SO}(n)$ *is a continuous surjective homomorphism.*

(b) $\ker \rho = \{\pm 1\}$.

(c) $\mathrm{Spin}(n)$ *is a compact connected Lie group, and ρ is a two-sheeted covering homomorphism.*

(d) For $n \geq 3$, using the cited fact that $\pi_1(\mathrm{SO}(n))$ has order two, the group $\mathrm{Spin}(n)$ is simply connected. So $\mathrm{Spin}(n)$ is the universal covering group of $\mathrm{SO}(n)$.

Proof. (a) Clearly $\rho(xy) = \rho(x)\rho(y)$. Since $x^{-1} = x^t$, the entries of $\rho(x)$ are polynomials in the coordinates of x. Given $g \in \mathrm{SO}(n)$, write $g = r_{w_1} \cdots r_{w_{2k}}$ with $2k \leq n$ by the Cartan–Dieudonné lemma, Lemma 13.3. By the reflection formula (13.2), $y = w_1 \cdots w_{2k}$ satisfies $\rho(y) = g$.

(b) Note $-1 = e_1 e_1 \in \mathrm{Spin}(n)$, and ± 1 act trivially. Conversely, let $\rho(x) = \mathrm{id}$, so x commutes with every e_i. Fix i. Sorting the basis elements of x by whether they contain e_i, and moving e_i to the front, gives

$$x = a + e_i b,$$

where a, b are combinations of e_J with $i \notin J$, a even and b odd. By the anticommutation relation (13.1), e_i commutes with a and anticommutes with b. Hence

$$xe_i = e_i a - e_i e_i b, \qquad e_i x = e_i a + e_i e_i b.$$

Equality forces $2e_i^2 b = -2b = 0$. So no basis element containing e_i occurs in x. As i was arbitrary, $x = \lambda 1$ with $\lambda \in \mathbb{R}$. Then $1 = x^t x = \lambda^2$, so $\lambda = \pm 1$.

(c) Let $N = 2\lfloor n/2 \rfloor \geq 2$ and $x \in \mathrm{Spin}(n)$. The surjectivity proof in part (a) gives $y = w_1 \cdots w_{2k}$ with $2k \leq N$ and $\rho(y) = \rho(x)$. Append pairs $e_1(-e_1) = 1$ until there are N factors. By the kernel statement (b), $x = \pm y$, so we may replace w_1 by $-w_1$ to get $x = y$. Hence $\mathrm{Spin}(n)$ is the image of the compact connected space $(S^{n-1})^N$ under the continuous map $(w_1, \ldots, w_N) \mapsto w_1 \cdots w_N$. So $\mathrm{Spin}(n)$ is compact and connected.

Being compact, the group $\mathrm{Spin}(n)$ is closed in $Cl(n)^\times$, hence a Lie group by the closed subgroup theorem (Lecture 3).

The smooth transitive action of $\mathrm{Spin}(n)$ on $\mathrm{SO}(n)$ through ρ has stabiliser $\{\pm 1\}$. The homogeneous-space theorem (Lecture 3) identifies $\mathrm{Spin}(n)/\{\pm 1\}$ with $\mathrm{SO}(n)$. The quotient by the discrete subgroup $\{\pm 1\}$ is a local diffeomorphism, so $d\rho$ is an isomorphism. The covering criterion of Lecture 4 applies and gives two sheets.

(d) We cite that $\pi_1(\mathrm{SO}(n))$ has order two for $n \geq 3$ (Hall, Chapter 13). So the universal covering $q\colon \tilde{G} \to \mathrm{SO}(n)$ has kernel of order two. Integrate $(d\rho)^{-1} dq$ by the integration theorem of Lecture 4 to a homomorphism $\phi\colon \tilde{G} \to \mathrm{Spin}(n)$. The homomorphisms $\rho \circ \phi$ and q have the same differential on connected $\tilde{G}$, so they agree.

Since dq and $d\rho$ are isomorphisms, so is $d\phi$. Thus $\phi(\tilde{G})$ is an open, hence closed, subgroup, and ϕ is onto by connectedness. Now $\ker q = \phi^{-1}(\{\pm 1\})$ has order $2|\ker \phi|$, so $\ker \phi$ is trivial. A bijective homomorphism of Lie groups is an isomorphism (Lecture 3). ∎

For $n = 2$, Exercise 13.1 shows that $\mathrm{Spin}(2)$ is a circle and ρ is $z \mapsto z^2$. So $\mathrm{Spin}(2)$ is not simply connected.

Example 13.5 (Rotations in a coordinate plane)**.** Fix $a \neq b$ and put

$$x_{ab}(t) = \cos t + \sin t\, e_a e_b = -(\cos t\, e_a + \sin t\, e_b)\, e_a \in \mathrm{Spin}(n).$$

Since $(e_a e_b)^2 = -1$, we have $x_{ab}(t) = \exp(t\, e_a e_b)$, so $e_a e_b \in \mathrm{Lie}(\mathrm{Spin}(n))$. The vectors e_a, e_b anticommute with $e_a e_b$, so $e_a x_{ab}(t)^{-1} = x_{ab}(t) e_a$, and similarly for e_b. Every e_c with $c \neq a, b$ commutes with $e_a e_b$. Using $e_a e_b e_a = e_b$ and $e_a e_b e_b = -e_a$ we get

$$\rho(x_{ab}(t))e_a = x_{ab}(2t)e_a = \cos 2t\, e_a + \sin 2t\, e_b, \qquad \rho(x_{ab}(t))e_b = -\sin 2t\, e_a + \cos 2t\, e_b,$$

and $\rho(x_{ab}(t))e_c = e_c$. So $\rho(x_{ab}(t))$ is the rotation through $2t$ in the (e_a, e_b)-plane. For $t \in [0, \pi]$ the rotations run once around a closed loop. The lift runs from 1 to $x_{ab}(\pi) = -1$, so it is not closed.

Since $\mathrm{Spin}(n) \subset Cl(n)_{\mathbb{C}}$, the spinor module gives a representation Π_S of $\mathrm{Spin}(n)$. The element $-1 \in \mathrm{Spin}(n)$ is the scalar -1 of $Cl(n)$. Hence

$$\Pi_S(-1) = -\mathrm{id}_S, \qquad \rho(-1) = \mathrm{id}\,.$$

So Π_S is not of the form $\Pi' \circ \rho$. This gives the opening sign change.

13.3 Spin(3), SU(2) and descent

In $Cl^0(3)$ put $\mathbf{i} = e_2e_3$, $\mathbf{j} = e_3e_1$, $\mathbf{k} = e_1e_2$. For instance $\mathbf{ij} = e_2e_3e_3e_1 = -e_2e_1 = \mathbf{k}$. In the same way

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1, \qquad \mathbf{ij} = -\mathbf{ji} = \mathbf{k}, \quad \mathbf{jk} = -\mathbf{kj} = \mathbf{i}, \quad \mathbf{ki} = -\mathbf{ik} = \mathbf{j}.$$

So $Cl^0(3) = \operatorname{span}\{1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is the quaternion algebra $\mathbb{H}$. Since $\mathbf{i}^t = e_3e_2 = -\mathbf{i}$, the transpose is quaternion conjugation $q \mapsto \bar{q}$. For $q = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$ we have $q\bar{q} = |q|^2 = a^2 + b^2 + c^2 + d^2$. Let $\mathrm{Sp}(1) = \{q \in \mathbb{H} : |q| = 1\}$ be the group of unit quaternions.

Define a real linear map $\Phi\colon \mathbb{H} \to M_2(\mathbb{C})$ by

$$\Phi(1) = I, \quad \Phi(\mathbf{i}) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \Phi(\mathbf{j}) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \Phi(\mathbf{k}) = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}.$$

Direct multiplication shows that these matrices satisfy the quaternion relations. For example, $\Phi(\mathbf{i})\Phi(\mathbf{j}) = \Phi(\mathbf{k})$. Hence Φ is an algebra homomorphism, and

$$\Phi(a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}) = \begin{pmatrix} a + bi & -c - di \\ c - di & a - bi \end{pmatrix}, \qquad \det \Phi(q) = |q|^2. \tag{13.3}$$

Proposition 13.6. *(a)* $\mathrm{Spin}(3) = \mathrm{Sp}(1)$.
(b) Φ *restricts to an isomorphism of Lie groups* $\mathrm{Sp}(1) \to \mathrm{SU}(2)$.

Proof. (a) If $q \in \mathrm{Spin}(3)$, then $q\bar{q} = qq^t = 1$ by the inverse formula of Lemma 13.2(a), so $|q| = 1$. Conversely, let $|q| = 1$ and write $q = a + p$ with $p \in \operatorname{span}\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$. If $p = 0$, then $q = \pm 1 \in \mathrm{Spin}(3)$. Otherwise put $w = p/|p|$. Cross terms in w^2 cancel by anticommutation, so $w^2 = -1$. Choose θ with $\cos\theta = a$ and $\sin\theta = |p|$. Then

$$q = \cos\theta + \sin\theta\, w = \exp(\theta w).$$

By the plane-rotation example, Example 13.5, $\mathbf{i}, \mathbf{j}, \mathbf{k} \in \mathrm{Lie}(\mathrm{Spin}(3))$. The Lie algebra of a closed subgroup is a linear subspace, and exp maps it into the subgroup (Lecture 3). So $\theta w \in \mathrm{Lie}(\mathrm{Spin}(3))$ and $q \in \mathrm{Spin}(3)$.

(b) By the matrix formula (13.3), $\Phi(\bar{q}) = \Phi(q)^*$. So $\Phi(q)\Phi(q)^* = \Phi(q\bar{q}) = I$ for $|q| = 1$, and $\det\Phi(q) = 1$. Conversely, let $g \in \mathrm{SU}(2)$ have first column $(\alpha, \beta)^T$. The second column is a unit vector orthogonal to the first, and $\det g = 1$. This forces it to be $(-\bar{\beta}, \bar{\alpha})^T$. So $g = \Phi(q)$ with $\alpha = a + bi$ and $\beta = c - di$. Thus Φ is a bijective continuous homomorphism. Its inverse is linear, hence continuous. ∎

From now on we identify Spin(3) with SU(2) through Φ. Then $\rho\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ has kernel $\{\pm I\}$.

This ρ is the cover Ad of Lecture 4, up to a rotation of coordinates. Indeed, Exercise 13.2(a) shows that ρ is conjugation in the basis $(\mathbf{i}, \mathbf{j}, \mathbf{k})$, and in the basis (X_1, X_2, X_3) of Lecture 4 we have $\Phi(\mathbf{i}) = -2X_3$, $\Phi(\mathbf{j}) = 2X_2$ and $\Phi(\mathbf{k}) = 2X_1$. So $\rho = O\ \mathrm{Ad}\ O^{-1}$, where O is the rotation with $Oe_1 = e_3$, $Oe_2 = e_2$ and $Oe_3 = -e_1$. Hence the diagonal matrices rotate about e_1 here, and about the third axis in Lecture 4 and Lecture 12. A spinor module of $Cl(3)_{\mathbb{C}}$ is $\mathbb{C}^2$ with

e_j acting by $i\sigma_j$, for the Pauli matrices σ_j of Lecture 4. It sends $\mathbf{i}, \mathbf{j}, \mathbf{k}$ to $-i\sigma_1, -i\sigma_2, -i\sigma_3$, and with this isomorphism in place of Φ, the map ρ is exactly Ad.

For $m \geq 0$ let V_m be the space of homogeneous polynomials of degree m in $z = (z_1, z_2)$, with

$$(\Pi_m(g)f)(z) = f(g^{-1}z) \qquad (g \in \mathrm{SU}(2)).$$

The space V_m has dimension $m+1$ and spin $m/2$. Every finite-dimensional irreducible representation of SU(2) is isomorphic to exactly one V_m (Lecture 8). Fix an SU(2)-invariant inner product on each V_m (Lecture 5).

The spinor module of $Cl(3)_{\mathbb{C}}$ has dimension two. If its restriction to SU(2) were reducible, complete reducibility would make it $V_0 \oplus V_0$, since V_0 is the only one-dimensional irreducible. That restriction would be trivial, contrary to $\Pi_S(-1) = -\mathrm{id}_S$. Thus the restriction is isomorphic to V_1.

By the Descent Theorem of Lecture 4, a representation Π of SU(2) has the form $\Pi' \circ \rho$ exactly when $\Pi(\pm I) = \mathrm{id}$. The hypotheses hold by Theorem 13.4.

Proposition 13.7. *Π_m factors through $\rho\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ if and only if m is even.*

Proof. We have $\ker\rho = \{\pm I\}$ and $(\Pi_m(-I)f)(z) = f(-z) = (-1)^m f(z)$. Apply the Descent Theorem of Lecture 4. ∎

So the irreducible representations of SO(3) are the descents of the V_{2l}, of dimension $2l+1$. The spinor V_1 is not among them.

Example 13.8 (A full turn on V_m)**.** Put

$$u(\theta) = \Phi(\cos\frac{\theta}{2} + \sin\frac{\theta}{2}\,\mathbf{i}) = \mathrm{diag}(e^{i\theta/2}, e^{-i\theta/2}).$$

Since $\mathbf{i} = e_2e_3$, the plane-rotation example, Example 13.5, shows that $\rho(u(\theta))$ is the rotation $R(\theta)$ through θ in the (e_2, e_3)-plane. On the monomial $f_k = z_1^{m-k}z_2^k$ we get

$$\Pi_m(u(\theta))f_k = (e^{-i\theta/2}z_1)^{m-k}(e^{i\theta/2}z_2)^k = e^{i(2k-m)\theta/2}f_k.$$

At $\theta = 2\pi$ every eigenvalue is $e^{i(2k-m)\pi} = (-1)^m$. Hence

$$R(2\pi) = \mathrm{id}, \qquad \Pi_m(u(2\pi)) = (-1)^m\,\mathrm{id}\,.$$

For odd m every vector changes sign. Each ray $\mathbb{C}f$ is unchanged, and so is every transition probability $|\langle f, h\rangle|^2/(\|f\|^2\|h\|^2)$. In physics, the sign is visible only relative to another part of a state, for example in the interference of a beam, one part of which has been turned through 2π.

Remark 13.9. Connectedness matters when a representation is recovered from its differential. The trivial character and det of O(2) both have differential 0 on $\mathfrak{so}(2)$, but they differ on a reflection. So for a disconnected group, $d\Pi$ need not determine Π (Exercise 13.4).

13.4 Projective unitary representations

In quantum mechanics a pure state is a ray $\mathbb{C}f$ in a Hilbert space, with $f \neq 0$. Transition probabilities depend only on rays. So a symmetry group acts on rays. Lifting this action to vectors may require choices of phase.

Let $\mathcal{H}$ be a complex Hilbert space of finite dimension $d \geq 1$, and let $\mathbb{T} = \{\lambda \in \mathbb{C} : |\lambda| = 1\}$.

Definition 13.10. The *projective unitary group* is $\mathrm{PU}(\mathcal{H}) = \mathrm{U}(\mathcal{H})/\mathbb{T}$ id with the quotient topology. We write $[U]$ for the class of U.

It is a Lie group (Lecture 3). It acts on rays by $[U] \cdot \mathbb{C}f = \mathbb{C}Uf$.

Definition 13.11. A *projective unitary representation* of a Lie group G on $\mathcal{H}$ is a continuous homomorphism $\bar{\pi} \colon G \to \mathrm{PU}(\mathcal{H})$.

We call $\bar{\pi}$ *irreducible* if 0 and $\mathcal{H}$ are the only subspaces invariant under every U with $[U] \in \bar{\pi}(G)$. A *lift* is any function $U \colon G \to \mathrm{U}(\mathcal{H})$ with $[U(g)] = \bar{\pi}(g)$. No continuity is assumed. Since $[U(g)U(h)] = [U(gh)]$, there is a function $c \colon G \times G \to \mathbb{T}$ with

$$U(g)U(h) = c(g,h)\,U(gh). \tag{13.4}$$

Definition 13.12. A *multiplier* of G is a function $c \colon G \times G \to \mathbb{T}$ such that

$$c(g,h)\,c(gh,k) = c(h,k)\,c(g,hk) \qquad (g,h,k \in G).$$

Multipliers c, c' are *equivalent* if $c'(g,h) = c(g,h)\,b(g)b(h)/b(gh)$ for some function $b \colon G \to \mathbb{T}$. A multiplier is *trivial* if it is equivalent to 1.

Lemma 13.13. *The function c in (13.4) is a multiplier. Its equivalence class does not depend on the lift. It is trivial if and only if $\bar{\pi}$ has a lift that is a homomorphism.*

Proof. Expand $U(g)U(h)U(k)$ in two ways:

$$(U(g)U(h))U(k) = c(g,h)c(gh,k)U(ghk), \qquad U(g)(U(h)U(k)) = c(h,k)c(g,hk)U(ghk).$$

This gives the identity. Any other lift is bU with $b \colon G \to \mathbb{T}$, and its multiplier is $c(g,h)\,b(g)b(h)/b(gh)$. So the class is well defined. The lift bU is a homomorphism exactly when this multiplier equals 1. ∎

Example 13.14 (Spin one half)**.** Let $\bar{\pi}_1$ be the action of $\mathrm{SO}(3)$ on rays of $\mathbb{C}^2$ of Corollary 13.18 below. For each rotation g choose one of its two preimages $s(g) \in \mathrm{SU}(2)$, and put $U(g) = \Pi_1(s(g))$. Since $s(g)s(h) = \pm s(gh)$ and $\Pi_1(-I) = -\mathrm{id}$, the multiplier c takes the values ± 1.

No choice of s is continuous. A continuous choice along the loop $t \mapsto \rho(x_{ab}(t))$, $t \in [0, \pi]$, would be a closed lift of this loop. Lifts are unique given their start (Lecture 4), and Example 13.5 shows that the lift starting at 1 runs to -1. Multiplying by -1 shows that the lift starting at -1 runs to 1. So no lift of the loop is closed.

Definition 13.15. A *central extension* of G by $\mathbb{T}$ is a Lie group E with a closed central subgroup $Z \cong \mathbb{T}$ and an isomorphism of Lie groups $E/Z \cong G$.

Proposition 13.16. *Let $\bar\pi$ be a projective unitary representation of G on $\mathcal{H}$. Then*

$$E_{\bar\pi} = \{(g, U) \in G \times \mathrm{U}(\mathcal{H}) : [U] = \bar\pi(g)\}$$

is a central extension of G by $Z = \{(1, \lambda\,\mathrm{id}) : \lambda \in \mathbb{T}\}$. The map $(g, U) \mapsto U$ is a unitary representation of $E_{\bar\pi}$ in which $(1, \lambda\,\mathrm{id})$ acts by λ.

Proof. The set $E_{\bar\pi}$ is a subgroup of $G \times \mathrm{U}(\mathcal{H})$. It is the preimage of the diagonal under the continuous map $(g, U) \mapsto (\bar\pi(g), [U])$ into the Hausdorff space $\mathrm{PU}(\mathcal{H})^2$. So it is closed, hence a Lie group (Lecture 3). Clearly Z is closed and central. The projection to G is surjective with kernel Z. The projection defines a smooth transitive action of $E_{\bar\pi}$ on G, with stabiliser Z. The homogeneous-space theorem of Lecture 3 therefore identifies $E_{\bar\pi}/Z$ smoothly with G. ∎

Conversely, a unitary representation of a central extension E in which Z acts by $\lambda \mapsto \lambda\,\mathrm{id}$ defines a projective representation of E/Z. In this sense, projective representations of G are ordinary representations of central extensions of G.

For $\bar\pi_1$ of Example 13.14, the map $(g, U) \mapsto U$ is an isomorphism $E_{\bar\pi_1} \to \mathrm{U}(2)$, with $Z = \mathbb{T}\,\mathrm{id}$, because $\bar\pi_1$ is a bijection $\mathrm{SO}(3) \to \mathrm{PU}(\mathbb{C}^2)$. So $\mathrm{U}(2)$ is a central extension of $\mathrm{SO}(3)$ by $\mathbb{T}$, and spin one half is an ordinary representation of it.

13.5 Lifting projective representations

Theorem 13.17. *Let G be a connected Lie group with universal covering homomorphism $q\colon \tilde G \to G$ (Lecture 4). Let $\bar\pi$ be a projective unitary representation of G on a finite-dimensional $\mathcal{H}$. Then there is a unique continuous homomorphism $U\colon \tilde G \to \mathrm{SU}(\mathcal{H})$ with $[U(x)] = \bar\pi(q(x))$.*

Proof. Let $d = \dim \mathcal{H}$. Every $V \in \mathrm{U}(\mathcal{H})$ equals λV_0 with $\lambda^d = \det V$ and $V_0 \in \mathrm{SU}(\mathcal{H})$. So the restriction $p\colon \mathrm{SU}(\mathcal{H}) \to \mathrm{PU}(\mathcal{H})$ of the quotient map is surjective. Its kernel $\{\lambda\,\mathrm{id} : \lambda^d = 1\}$ is finite, and $\mathrm{SU}(\mathcal{H})$ is connected.

The differential of p has kernel $\mathfrak{su}(d) \cap i\mathbb{R}I = 0$. Both $\mathfrak{su}(d)$ and the tangent algebra of $\mathrm{U}(d)/\mathbb{T}I$ have dimension $d^2 - 1$, so dp is an isomorphism. By the covering criterion of Lecture 4, p is a covering homomorphism. For $d = 1$ both groups are trivial.

Since $\tilde G$ is simply connected, the based lifting theorem of Lecture 4 gives a unique continuous lift U with $U(e) = I$ and $p \circ U = \bar\pi \circ q$. It remains to check that U is multiplicative. The two maps $(x, y) \mapsto U(xy)$ and $(x, y) \mapsto U(x)U(y)$ lift the same map on the connected space $\tilde G \times \tilde G$ and agree at (e, e). Uniqueness of covering lifts makes them equal. Thus U is a homomorphism. Any homomorphic lift sends e to I, so uniqueness of the based lift also gives the stated uniqueness. ∎

Corollary 13.18. *Let $\mathcal{H}$ be finite-dimensional.*

(a) For each projective unitary representation $\bar{\pi}$ of $\mathrm{SO}(3)$ on $\mathcal{H}$, there is a unitary representation U of $\mathrm{SU}(2)$ with $\bar{\pi}(\rho(x)) = [U(x)]$. Moreover $U(-I) = \pm\,\mathrm{id}$.

(b) If $\bar{\pi}$ is irreducible, then U is unitarily equivalent to Π_m with $m = \dim \mathcal{H} - 1$.

(c) For each $m \geq 0$, the rule $\bar{\pi}_m(\rho(x)) = [\Pi_m(x)]$ defines a projective unitary representation of $\mathrm{SO}(3)$. Its multiplier is trivial if and only if m is even.

Proof. (a) The group $\mathrm{SU}(2) \cong S^3$ is simply connected (Lecture 4). Apply the lifting theorem, Theorem 13.17. Since $[U(-I)] = \bar{\pi}(\mathrm{id})$, $U(-I)$ is a scalar. Its square is $U(I) = \mathrm{id}$, so the scalar is ± 1.

(b) Scalars preserve every subspace, so U is irreducible. By the classification, $U \cong \Pi_m$ with $m + 1 = \dim \mathcal{H}$. By Schur's lemma the isomorphism becomes unitary after rescaling (Lecture 5).

(c) If $\rho(x) = \rho(x')$, then $x' = \pm x$ and $\Pi_m(x') = \pm \Pi_m(x)$. So $\bar{\pi}_m$ is well defined. It is continuous by the Descent Theorem of Lecture 4, with target $\mathrm{PU}(V_m)$. For even m, the descent criterion Proposition 13.7 gives a lift that is a homomorphism.

Let m be odd. Put $A = \rho(\Phi(\mathbf{i}))$ and $B = \rho(\Phi(\mathbf{j}))$. Since $\mathbf{i} = x_{23}(\pi/2)$ and $\mathbf{j} = x_{31}(\pi/2)$, the plane-rotation example, Example 13.5, gives

$$A = \mathrm{diag}(1, -1, -1), \qquad B = \mathrm{diag}(-1, 1, -1), \qquad AB = BA.$$

Put $P = \Pi_m(\Phi(\mathbf{i}))$ and $Q = \Pi_m(\Phi(\mathbf{j}))$. Since $\mathbf{ij} = -\mathbf{ji}$ and $\Pi_m(-I) = -\,\mathrm{id}$,

$$PQ = \Pi_m(-I)\, QP = -QP.$$

Suppose $W\colon\ \mathrm{SO}(3) \to \mathrm{U}(V_m)$ is a homomorphism lifting $\bar{\pi}_m$, with no continuity assumed. Then $W(A) = \alpha P$ and $W(B) = \beta Q$ with $\alpha, \beta \in \mathbb{T}$. So $W(A)W(B) = -W(B)W(A)$. But $AB = BA$ gives $W(A)W(B) = W(B)W(A)$. Hence $2W(A)W(B) = 0$, which is impossible for invertible operators. By the triviality criterion of Lemma 13.13, the multiplier is nontrivial. ∎

So every finite-dimensional irreducible projective representation of $\mathrm{SO}(3)$ is equivalent to some $\bar{\pi}_m$. Here equivalence means that a unitary $T : V_m \to \mathcal{H}$ satisfies

$$\bar{\pi}(\rho(x)) = [T \Pi_m(x) T^{-1}] \qquad (x \in \mathrm{SU}(2)).$$

All spins $m/2$ occur, but only integer spins come from genuine representations. This is the mathematical form of the half-integer spin of the electron.

For separable Hilbert spaces we cite the following result.

Theorem 13.19 (Bargmann; cited)**.** *Let G be a connected Lie group with semisimple Lie algebra $\mathfrak{g}$ and universal covering $q\colon \tilde{G} \to G$. Let $\mathcal{H}$ be a separable Hilbert space, and let $\bar{\pi}$ be a homomorphism from G to $\mathrm{U}(\mathcal{H})/\mathbb{T}\,\mathrm{id}$. Assume that $g \mapsto |\langle f, U(g)h\rangle|$ is continuous for all $f, h \in \mathcal{H}$ and one (hence every) lift U. Then there is a strongly continuous unitary representation $\tilde{U}$ of $\tilde{G}$ with $[\tilde{U}(x)] = \bar{\pi}(q(x))$ for $x \in \tilde{G}$.*

Example 13.20 ($\mathbb{R}^2$ and the Heisenberg extension)**.** On $G = \mathbb{R}^2$ put

$$\omega(x, y) = x_1 y_2 - x_2 y_1, \qquad c(x, y) = e^{i\omega(x,y)/2}.$$

The exponent is bilinear, so c is a multiplier. The factor $b(x)b(y)/b(x+y)$ is symmetric in x, y. So the ratio

$$c(x, y)/c(y, x) = e^{i\omega(x,y)}$$

depends only on the class of c. The ratio is $e^i \neq 1$ for $x = (1, 0)$ and $y = (0, 1)$, so c is nontrivial. Yet $\mathbb{R}^2$ is its own universal cover.

On $E = \mathbb{R}^2 \times \mathbb{T}$ put $(x, \lambda)(y, \mu) = (x+y, \lambda\mu\, c(x, y))$. The multiplier identity gives associativity, and Exercise 13.5 shows that E is a group. Its multiplication is smooth. Projection onto $\mathbb{R}^2$ identifies the quotient $E/(\{0\} \times \mathbb{T})$ with $\mathbb{R}^2$, so E is a central extension. The commutator of (x, λ) and (y, μ) in E is $(0, e^{i\omega(x,y)})$.

This does not contradict the lifting theorem, Theorem 13.17. A lift on $\mathbb{C}^d$ with multiplier equivalent to c would satisfy $U(x)U(y) = e^{i\omega(x,y)}U(y)U(x)$. Taking determinants gives $e^{id\,\omega(x,y)} = 1$ for all x, y, which is false. So c occurs only on infinite-dimensional spaces (Lecture 18). In physics, this is the group form of the fact that $\operatorname{tr}[x, p] = 0$ rules out $[x, p] = i\hbar$ on a finite-dimensional space. Bargmann's theorem does not apply to $\mathbb{R}^2$, because its Lie algebra is abelian rather than semisimple.

Self-check questions

1. In Spin(4), what is $\rho(e_1e_2e_3e_4)$?
   Answer. The element $e_1e_2e_3e_4$ equals $x_{12}(\pi/2)\, x_{34}(\pi/2)$. By the plane-rotation example, Example 13.5, its image under ρ rotates through π in both coordinate planes, so $\rho(e_1e_2e_3e_4) = -\mathrm{id}$. The other lift of $-\mathrm{id}$ is $-e_1e_2e_3e_4$.
2. For odd m, does $\Pi_m(u(2\pi))$ change any transition probability?
   Answer. No. It equals $-\mathrm{id}$. Hence it fixes every ray, and $|\langle -f, -h\rangle| = |\langle f, h\rangle|$.
3. Is there a projective unitary representation of $\mathbb{R}^2$ on $\mathbb{C}^3$ with multiplier $e^{i\omega/2}$?
   Answer. No. The determinant argument of the Heisenberg example, Example 13.20, with $d = 3$ gives $e^{3i\omega(x,y)} = 1$ for all x, y, which is false.

Summary

The group $\mathrm{Spin}(n)$ consists of even products of unit vectors in $Cl(n)$. For $n \geq 2$, conjugation maps it onto $\mathrm{SO}(n)$ with kernel $\{\pm 1\}$. For $n \geq 3$, the cited value of $\pi_1(\mathrm{SO}(n))$ shows that it is the universal cover. Spinor modules send -1 to $-\mathrm{id}$ and do not factor through ρ.

For $n = 3$, $\mathrm{Spin}(3) = \mathrm{Sp}(1) \cong \mathrm{SU}(2)$, and V_m descends to $\mathrm{SO}(3)$ exactly for even m. The lift $u(2\pi)$ of a full turn acts on V_m by $(-1)^m$, which fixes every ray.

In finite dimensions, projective unitary representations of connected Lie groups lift to unitary representations of the universal cover. On separable Hilbert spaces, Bargmann's cited theorem gives strongly continuous lifts for connected groups with semisimple Lie algebra, under its continuity hypothesis. The multiplier on the simply connected group $\mathbb{R}^2$ shows that semisimplicity cannot be dropped.

Reading

- BD, Section I.6: Clifford algebras and Spin(n). Note the sign convention for v^2.
- Sepanski, Section 1.3: the Clifford-algebra construction of the double cover of SO(n).
- Kirillov, Elements, §14: projective representations and their multipliers.
- Hall, Chapter 13: fundamental groups of compact Lie groups, including SO(n).
- V. Bargmann, On unitary ray representations of continuous groups, Ann. of Math. 59 (1954).